Valentinos Pariza

PhD Researcher in Computer Vision · University of Technology Nuremberg
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Research Profile

My research connects how models understand images with how they generate and edit them. Through NeCo and Franca, I worked on self-supervised vision encoders and the spatial information in their representations. I now study diffusion models for image editing, generation, and visual perception, with growing interests in multimodal learning and video. I want to understand what visual knowledge transfers between tasks, and how to use it to build more capable models through efficient training and adaptation.

Education

University of Technology Nuremberg (UTN) Dec 2024–present
PhD research in Deep Learning for Vision, Fundamental AI Lab *Nuremberg, Germany*

Advisor: Prof. Yuki M. Asano.

University of Amsterdam Sep 2022–Aug 2024
MSc in Artificial Intelligence *Amsterdam, Netherlands*

GPA: 9.03/10; Graduated cum laude. ELLIS MSc Honours Programme.
Thesis research: improving and evaluating dense visual representations (NeCo and Open Hummingbird Evaluation).
Thesis supervisors: Yuki M. Asano (UvA), Gertjan Burghouts (TNO), and Francesco Locatello (ISTA).

University of Cyprus Sep 2017–Jun 2022
BSc in Computer Science, AI specialization *Nicosia, Cyprus*

GPA: 9.36/10; second-highest GPA in the graduating Computer Science class.
Thesis Research: on accelerating quality-diversity optimization for robot tasks using JAX and differentiability.

Publications

Franca: Nested Matryoshka Clustering for Scalable Visual Representation Learning.
S. Venkataramanan*, **V. Pariza***, M. Salehi, L. Knobel, E. Ramzi, S. Gidaris, A. Bursuc, Y. M. Asano. **CVPR 2026**. *Equal contribution. [Paper]

Near, far: Patch-ordering enhances vision foundation models' scene understanding (NeCo).
V. Pariza, M. Salehi, G. Burghouts, F. Locatello, Y. M. Asano. **ICLR 2025**. [Paper]

[Re] Reproducibility Study of "Label-Free Explainability for Unsupervised Models".
V. Pariza, A. Pal, M. Pawar, Q. S. Faber. **ReScience C**, 9(2), Article 11, 2023. [Paper]

Research Experience

PhD Researcher Dec 2024–present
Fundamental AI Lab, UTN *Nuremberg, Germany*

- **Open Vision Banana (ongoing; manuscript in preparation):** Core contributor to adapting image editors for perception tasks, like depth estimation and segmentation. Work: mainly on depth and surface-normal prediction, image editing, and generation. Public release planned.
- **Franca:** Co-first author and core contributor to an open vision foundation model competitive with DINOv2 on dense tasks. Work: ideas, code, standard- and high-resolution training, experiments, ablations, and writing.
- **RASA (part of Franca):** Designed and implemented RASA to remove unwanted absolute-position information from visual features. Work: Full contribution, including running its experiments and ablations.

Honors and Awards

Outstanding Reviewer, CVPR 2026
Best Ready for Production Solution, Summer of AI, South Pole project 2023
1st place, Go Green, Go Digital, Go Start-up! competition, Fooderloo 2021
Best & Most Innovative Solution, Cyprus Digital Championship for Students 2020

Earlier Research Experience

Research Intern Feb–Jul 2024
TNO — Netherlands Organisation for Applied Scientific Research *Netherlands*

- Developed NeCo with collaborators during my MSc thesis (ICLR 2025). In the reported setup, 19 hours of post-training on a single GPU improved dense visual representations, by up to ~7 percentage points in evaluations.
- Developed and released Open Hummingbird Evaluation, an implementation of Balažević et al.'s dense nearest-neighbor evaluation for vision encoders.

Research Intern Jun–Jul 2022
CYENS Centre of Excellence *Nicosia, Cyprus*

- Built a Python library for quality-diversity optimization in JAX, more than doubling the speed of the algorithms tested.

Selected Additional Research/Software Projects

Robustness of 3D occupancy prediction · UvA course project 2023
Tested 3D occupancy models with synthetically corrupted inputs and studied how data augmentation affects performance on clean and corrupted data. [Code]

Socratic Image Generation · UvA course project 2023
Built a pipeline that uses image captions to refine image-generation prompts over several rounds. Studied bias and hallucinations when combining the models. [Code]

Industry and Other Experience

Data Science Intern Jul–Aug 2023
South Pole *Amsterdam, Netherlands*

- Worked with other teams to understand where product matching went wrong, what they needed, and which tools we could reuse.
- Built an evaluation dataset and used a text transformer encoder to improve product-to-category matching accuracy by **25.2 percentage points**.
- Deployed the models to production using Kubernetes.

Software Development Engineer Intern Aug 2020–Jul 2021
Amazon Data Services Ireland *Dublin, Ireland*

- Spoke with other teams about reporting and logging problems, identified bottlenecks, and used their requirements to guide software improvements.
- Built and deployed a reporting service that saved each employee involved about 8–10 hours of manual reporting per week. Along this work, improved a Java library for accessing internal data.
- Improved a real-time log analysis service and led two campaigns to improve logging practices, reducing annual costs by thousands of dollars.

Co-Founder & Backend Developer Oct 2019–Mar 2022
Fooderloo *Cyprus*

Co-founded Fooderloo to help reduce food waste in Cyprus. Built the backend and database.

Research and Technical Skills

Programming: Python, C/C++, Java, SQL.
Deep learning: PyTorch, PyTorch Lightning, JAX, TensorFlow; Hugging Face Transformers and Diffusers.
GPU training: Multi-GPU training (DDP/FSDP), mixed precision, profiling, memory optimization, Slurm.
Research methods: Experimental design, ablations, robustness testing, reproducibility and model evaluation.
Development and deployment: Git, Docker, Kubernetes, AWS, Google Cloud Platform.